Ex # 13.5

Date: _____

Chain Rule :-

$z = z(x,y)$, $x = x(t)$, $y = y(t)$

$\frac{dz}{dt} = ?$ $\quad \frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$

$y = t^2$, $x = t^3$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{3t^2} = \frac{2}{3}t.$$

1-6 Use an appropriate form of chain rule to find dz/dt.

1. $z = 3x^2y^3$; $x = t^4$, $y = t^2$

Solu:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}$$

$$= 6xy^3(4t^3) + 9x^2y^2(2t)$$

$$= 6(t^4)(t^6)(4t^3) + 9(t^8)(t^4)(2t$$

$$= 24t^{13} + 18t^{13} = 42t^{13}$$

2. $z = \ln(2x^2 + y)$; $x = \sqrt{t}$, $y = t^{2/3}$

Solu:

We find the necessary derivatives.

$\cdot \frac{\partial z}{\partial x} = \frac{4x}{2x^2+y}$ $\quad \frac{\partial z}{\partial y} = \frac{1}{2x^2+y}$

$\cdot \frac{dx}{dt} = \frac{1}{2}t^{-1/2}$ $\quad \frac{dy}{dt} = \frac{2}{3}t^{-1/3}$

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Apply the chain rule and substitute for x and y.

$$\frac{dz}{dt} = \left(\frac{4x}{2x^2+y}\right)\left(\frac{1}{2}t^{-1/2}\right) + \left(\frac{1}{2x^2+y}\right)\left(\frac{2}{3}t^{-1/3}\right)$$

$$\frac{dz}{dt} = \left(\frac{4\sqrt{t}}{2t+t^{2/3}}\right)\left(\frac{1}{2\sqrt{t}}\right) + \left(\frac{1}{2t+t^{2/3}}\right)\left(\frac{2}{3}t^{-1/3}\right)$$

$$\frac{dz}{dt} = \frac{2}{2t+t^{2/3}} + \frac{\frac{2}{3}t^{-1/3}}{2t+t^{2/3}} = \frac{2+\frac{2}{3}t^{-1/3}}{2t+t^{2/3}}$$

$$\frac{dz}{dt} = \frac{\frac{2}{3}(3+t^{-1/3})}{2t+t^{2/3}} = \frac{2(3+t^{-1/3})}{3(2t+t^{2/3})}$$ Ans

3. $z = 3\cos x - \sin xy$; $x = 1/t$, $y = 3t$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}$$

$$\frac{\partial z}{\partial x} = -3\sin x - y\cos(xy)$$

$$\frac{dz}{\partial y} = -x\cos(xy)$$

$$dx/dt = -1/t^2$$

$$dy/dt = 3$$

Applying the chain rule. Note that $xy = (1/t)(3t) = 3$.

$$\frac{dz}{dt} = (-3\sin x - y\cos(xy))\left(-\frac{1}{t^2}\right) + (-x\cos(xy))(3)$$

$$\frac{dz}{dt} = \left(-3\sin\left(\frac{1}{t}\right) - 3t\cos(3)\right)\left(-\frac{1}{t^2}\right) + \left(\frac{1}{t}\cos(3)\right)(3)$$

$$\frac{dz}{dt} = \frac{3}{t^2}\sin\left(\frac{1}{t}\right) + \frac{3t}{t^2}\cos(3) - \frac{3}{t}\cos(3).$$

$$\frac{dz}{dt} = 3t^{-2}\sin(1/t).$$ Ans

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4. $z = 3\cos x - \sin xy$; $x = \ln t$, $y = t$

We find the necessary derivatives:

$$\frac{\partial z}{\partial x} = \frac{1 - 2y^4}{2\sqrt{1 + x - 2xy^4}}$$

$$\frac{\partial z}{\partial y} = \frac{-4xy^3}{\sqrt{1 + x - 2xy^4}}$$

$$\frac{dx}{dt} = \frac{1}{t}$$

$$\frac{dy}{dt} = 1$$

Apply the chain rule:

$$\frac{dz}{dt} = \left(\frac{1 - 2y^4}{2\sqrt{1 + x - 2xy}}\right)\left(\frac{1}{t}\right) - \frac{4xy^3}{\sqrt{1 + x - 2xy^4}}$$

$$\frac{dz}{dt} = \frac{1 - 2t^4 - 4t(\ln t)(t^3)}{2t\sqrt{1 + \ln t - 2(\ln t)t^4}} = \frac{1 - 2t^4 - 4t^4 \ln t}{2t\sqrt{1 + \ln t - 2t^4 \ln t}}$$

5. $z = e^{1-xy}$; $x = t^{1/3}$, $y = t^3$

We find the necessary derivatives:

$$\frac{\partial z}{\partial x} = -ye^{1-xy}$$

$$\frac{\partial z}{\partial y} = -xe^{1-xy}$$

$$\frac{dx}{dt} = \frac{1}{3}t^{-2/3}$$

$$\frac{dy}{dt} = 3t^2$$

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Apply the chain rule.

$$\frac{dz}{dt} = \left(-ye^{1-xy}\right)\left(\frac{1}{3}t^{-2/3}\right) + \left(-xe^{1-xy}\right)\left(3t^2\right)$$

$$= e^{1-xy}\left(-\frac{1}{3}yt^{-2/3} - 3xt^2\right)$$

$$= e^{1-t^{10/3}}\left(-\frac{1}{3}(t^3)t^{-2/3} - 3(t^{1/3})t^2\right)$$

$$= e^{1-t^{10/3}}\left(-\frac{1}{3}t^{7/3} - 3t^{7/3}\right) = e^{1-t^{10/3}}\left(-\frac{10}{3}t^{7/3}\right)$$

$$= -\frac{10}{3}t^{7/3}e^{1-t^{10/3}}$$ Ans

6. $z = \cosh^2 xy$; $x = t/2$, $y = e^t$

Solution

$$\frac{\partial z}{\partial x} = 2\cosh(xy)\cdot\sinh(xy)\cdot y = 2y\cosh(xy)\sinh(xy)$$

$$\frac{\partial z}{\partial y} = 2\cosh(xy)\cdot\sinh(xy)\cdot x = 2x\cosh(xy)\sinh(xy)$$

$$\frac{dx}{dt} = \frac{1}{2}, \quad \frac{dy}{dt} = e^t$$

Applying the chain rule.

$$\frac{dz}{dt} = \left(2y\cosh(xy)\sinh(xy)\right)\left(\frac{1}{2}\right) + \left(2x\cosh(xy)\sinh(xy)\right)\left(e^t\right)$$

$$= \cosh(xy)\sinh(xy)\left[y + 2xe^t\right]$$

Substituting the value of x and y.

$$\frac{dz}{dt} = \cosh\left(\frac{te^t}{2}\right)\sinh\left(\frac{te^t}{2}\right)\left[e^t + 2\left(\frac{t}{2}\right)e^t\right]$$

$$= (1+t)e^t\cosh\left(\frac{te^t}{2}\right)\sinh\left(\frac{te^t}{2}\right)$$ Ans

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7-10 Use an appropriate form of the chain rule to find dw/dt.

7. $w = 5x^2y^3z^4$; $x = t^2$, $y = t^3$, $z = t^5$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$$

$$\frac{dw}{dt} = (10xy^3z^4)(2t) + (15x^2y^2z^4)(3t^2) + (20x^2y^3z^3)(5t^4)$$

$$= (10(t^2)(t^3)^3(t^5)^4(2t) + (15(t^2)^2(t^3)^2(t^5)^4)(3t^2) + (20(t^2)^2(t^3)^3(t^5)^3)(5t^4)$$

$$= (10t^{31})(2t) + (15t^{30})(3t^2) + (20t^{28})(5t^4)$$

$$= 20t^{32} + 45t^{32} + 100^{32}$$

$$= 165t^{32}$$

8. $w = \ln(3x^2 - 2y + 4z^3)$; $x = t^{1/2}$, $y = t^{2/3}$, $z = t^{-2}$

$$\frac{dw}{dt} = \left(\frac{6x}{3x^2 - 2y + 4x^2}\right)\left(\frac{1}{2}t^{-1/2}\right) + \left(\frac{-2}{3x^2 - 2y + 4x^2}\right)\left(\frac{2}{3}t^{-2/3}\right) + \left(\frac{12z^2}{3x^2 - 2y + 4x^2}\right)(-2t^{-3})$$

$$= \frac{1}{3t - 2t^{2/3} + 4t^{-6}}\left[(6t^{1/2})\left(\frac{1}{2}t^{-1/2}\right) - \frac{4}{3}t^{-1/3} + (12(t^{-2})^2)(-2t^{-3})\right.$$

$$= \frac{1}{3t - 2t^{2/3} + 4t^{-6}}\left[3 - \frac{4}{3}t^{-1/3} - 24t^{-7}\right]$$

$$= \frac{3 - \frac{4}{3}t^{-1/3} - 24t^{-7}}{3t - 2t^{2/3} + 4t^{-6}}$$

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9. $w = 5\cos xy - \sin xz$; $x = 1/t$, $y = t$, $z = t^3$

Solution

Note that $xy = 1$ and $xz = t^2$.

$$\frac{dw}{dt} = \left(-5y\sin(xy) - z\cos(xz)\right)\left(-\frac{1}{t^2}\right) + \left(-5x\sin(xy)\right)(1) + \left(-x\cos(xz)\right)(3t^2)$$

$$= \left(-5t\sin(1) - t^3\cos(t^2)\right)\left(-\frac{1}{t^2}\right) - \frac{5}{t}\sin(1) - \frac{1}{t}\cos(t^2)(3t^2)$$

$$= \left(\frac{5}{t}\sin(1) + t\cos(t^2)\right) - \frac{5}{t}\sin(1) - 3t\cos(t^2)$$

$$= t\cos(t^2) - 3t\cos(t^2) = -2t\cos(t^2)$$ ans

10. $w = \sqrt{1 + x - 2yz^4x}$; $x = \ln t$, $y = t$, $z = 4t$.

Solution

$$\frac{dw}{dw} = \left(\frac{1 - 2yz^4}{2\sqrt{\cdots}}\right)\left(\frac{1}{t}\right) + \left(\frac{-2xz^4}{2\sqrt{\cdots}}\right)(1) + \left(\frac{-8xyz^3}{2\sqrt{\cdots}}\right)(4)$$

$$= \frac{(1 - 2yz^4) - 2txz^4 - 32txyz^3}{2t\sqrt{1 + x - 2yz^4x}}$$

$$= \frac{1 - 2(t)(4t)^4 - 2t(\ln t)(4t)^4 - 32(\ln t)(t)(4t)^3}{2t\sqrt{1 + \ln t - 2(t)(4t)^4(\ln t)}}$$

$$= \frac{1 - 512t^5 - 512t^5\ln t - 2048t^5\ln t}{2t\sqrt{1 + \ln t - 512t^5\ln t}}$$

$$= \frac{1 - 512t^5 - 2560t^5\ln t}{2t\sqrt{1 + \ln t - 512t^5\ln t}}$$ ans

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17-22 Use appropriate forms of the chain rule to find $\partial z/\partial u$ and $\partial z/\partial v$.

17. $z = 8x^2y - 2x + 3y$; $x = uv$, $y = u - v$.

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial u} = (16xy - 2)(v) + (8x^2 + 3)(1)$$

$$= (16(uv)(u-v) - 2)v + (8(uv)^2 + 3)$$

$$= (16u^2v^2 - 16uv^3 - 2v) + 8u^2v^2 + 3$$

$$= 24u^2v^2 - 16uv^3 - 2v + 3.$$

$$\frac{\partial z}{\partial v} = (16xy - 2)(u) + (8x^2 + 3)(-1)$$

$$= (16(uv)(u-v) - 2)u - (8(uv)^2 + 3)$$

$$= (16u^2v - 16uv^2 - 2)u - 8u^2v^2 - 3$$

$$= 16u^3v - 16u^2v^2 - 2u - 8u^2v^2 - 3$$

$$= 16u^3v - 24u^2v^2 - 2u - 3$$

18. $z = x^2 - y\tan x$; $x = u/v$, $y = u^2v^2$

$$\frac{\partial z}{\partial u} = (2x - y\sec^2(x))\left(\frac{1}{v}\right) + (-\tan(x))(2uv^2)$$

$$= \left(\frac{2u}{v} - u^2v^2\sec^2\left(\frac{u}{v}\right)\right)\frac{1}{v} - 2uv^2\tan\left(\frac{u}{v}\right)$$

$$= \frac{2u}{v^2} - u^2v\sec^2\left(\frac{u}{v}\right) - 2uv^2\tan\left(\frac{u}{v}\right)$$

$$\frac{\partial z}{\partial v} = (2x - y\sec^2(x))\left(-\frac{u}{v^2}\right) + (-\tan(x))(2u^2v)$$

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$$= \left(\frac{2u}{v} - u^2v^2\sec\left(\frac{u}{v}\right)\right)\left(-\frac{u}{v^2}\right) - 2u^2v\tan\left(\frac{u}{v}\right)$$

$$= -\frac{2u^2}{v^3} + u^3\sec^2\left(\frac{u}{v}\right) - 2u^2v\tan\left(\frac{u}{v}\right)$$ Ans

19. $z = x/y$; $x = 2\cos u$, $y = 3\sin v$

Solve

$$\frac{\partial z}{\partial u} = \left(\frac{1}{y}\right)(-2\sin u) + \left(-\frac{x}{y^2}\right)(0)$$

$$= \frac{1}{3\sin v}(-2\sin u) = -\frac{2\sin u}{3\sin v}$$ Ans

$$\frac{\partial z}{\partial v} = \left(\frac{1}{y}\right)(0) + \left(-\frac{x}{y^2}\right)(3\cos v)$$

$$= \left(-\frac{2\cos u}{(3\sin v)^2}\right)(3\cos v) = -\frac{2\cos u\cos v}{3\sin^2 v}$$ Ans

20. $z = 3x - 2y$; $x = u + v\ln u$, $y = u^2 - v\ln v$

Solve

$$\frac{\partial z}{\partial u} = (3)\left(1 + \frac{v}{u}\right) + (-2)(2u) = 3 + \frac{3v}{u} - 4u$$ Ans

$$\frac{\partial z}{\partial v} = (3)(\ln u) + (-2)(-\ln v - 1)$$

$$= 3\ln u + 2\ln v + 2$$

$$= 2 + 3\ln u + 2\ln v$$ Ans

23-30 Use appropriate forms of the chain rule to find the derivatives.

23. Let $T = x^2y - xy^3 + 2$; $x = r\cos\theta$, $y = r\sin\theta$. Find $\partial T/\partial r$ and $\partial T/\partial\theta$.

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Sol:

$$\frac{\partial T}{\partial r} = (2xy - y^3)(\cos\theta) + (x^2 - 3xy^2)(\sin\theta)$$

$$= (2(r^2\cos\theta\sin\theta) - r^3\sin^3\theta)(\cos\theta) + (r^2\cos^2\theta - 3(r^3\cos\theta\sin^2\theta))(\sin\theta).$$

$$= (2r^2\cos^2\theta\sin\theta - r^3\sin^3\theta\cos\theta) + (r^2\cos^2\theta\sin\theta - 3r^3\cos\theta\sin^3\theta)$$

$$= 3r^2\sin\theta\cos^2\theta - 4r^3\sin^3\theta\cos\theta.$$ Ans.

$$\frac{\partial T}{\partial \theta} = (2xy - y^3)(-r\sin\theta) + (x^2 - 3xy^2)(r\cos\theta)$$

$$= (-2r^3\cos\theta\sin^2\theta + r^4\sin^4\theta) + (r^3\cos^3\theta - 3r^4\cos^2\theta\sin^2\theta)$$

$$= -2r^3\sin^2\theta\cos\theta + r^4\sin^4\theta + r^3\cos^3\theta - 3r^4\sin^2\theta\cos^2\theta.$$ Ans.

24. Let $R = e^{25-t^2}$; $S = 3\phi$, $t = \phi^{1/2}$. Find $dR/d\phi$.

Sol:

$$\frac{dR}{d\phi} = (2e^{25-t^2})(3) + (-2te^{25-t^2})\left(\frac{1}{2}\phi^{-\frac{1}{2}}\right)$$

$$\frac{dR}{d\phi} = 6e^{5\phi} - (\phi^{1/2}e^{5\phi})\left(\frac{1}{2}\phi^{-1/2}\right)$$

$$\frac{dR}{d\phi} = 6e^{5\phi} - e^{5\phi} = 5e^{5\phi}$$ Ans.

41-44 Use Theorem 13.5.3 to find dy/dx and check your result using implicit differentiation.

41. $x^2y^3 + \cos y = 0$

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$$\frac{d}{dx}(x^2y^3) + \frac{d}{dx}\cos y = 0$$

$$2xy^3 + x^2 3y^2 y' - \sin y\, y' = 0$$

$$y'(3x^2y^2 - \sin y) + 2xy^3 = 0$$

$$y' = \frac{-2xy^3}{3x^2y^2 - \sin y}$$ Ans

42. $x^3 - 3xy^2 + y^3 = 5$

solve

$$3x^2 - 3(y^2 + x\,2y\,y') + 3y^2y' = 0$$

$$3x^2 - 3y^2 - 6xyy' + 3y'y = 0$$

$$y'(3y^2 - 6xy) + (3x^2 - 3y^2) = 0$$

$$y'(3y^2 - 6xy) = 3y^2 - 3x^2$$

$$y' = \frac{3y^2 - 3x^2}{3y^2 - 6xy} = \frac{y^2 - x^2}{y - 2xy}$$

43. $e^{xy} + ye^y = 1$

solve

$$\frac{d}{dx}e^{xy} + \frac{d}{dx}(ye^y) = 0$$

$$e^{xy}(y + xy') + (y'e^y + ye^y y') = 0$$

$$e^{xy}y + y'(xe^{xy} + e^y(1+y)) = 0$$

$$y' = \frac{-e^{xy}y}{xe^{xy} + (1+y)e^y} = -\frac{ye^{xy}}{xe^{xy} + ye^y + e^y}$$ Ans

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44. $x - \sqrt{xy} + 3y = 4$

Solve

$$1 - \frac{1}{2\sqrt{xy}} \cdot (y + xy') + 3y' = 0$$

$$1 - \frac{y + xy' + 3y'}{2\sqrt{xy}} = 0$$

$$1 - \frac{y}{2\sqrt{xy}} - \frac{xy'}{2\sqrt{xy}} + 3y' = 0$$

$$y'\left(3 - \frac{x}{2\sqrt{xy}}\right) + 1 - \frac{y}{2\sqrt{xy}} = 0$$

$$y'\left(3 - \frac{x}{2\sqrt{xy}}\right) = \frac{y}{2\sqrt{xy}} - 1$$

$$y' = \frac{\frac{y}{2\sqrt{xy}} - 1}{3 - \frac{x}{2\sqrt{xy}}} = \frac{y - 2\sqrt{xy}}{6\sqrt{xy} - x}$$

Ans

Ex # 13.6

1-8 Find $D_u f$ at P.

1. $f(x,y) = (1+xy)^{3/2}$; $P(3,1)$; $u = \frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$

Solve

Gradient:

$$\nabla f(x,y) = \frac{3y}{2}\sqrt{1+xy}\, i + \frac{3x}{2}\sqrt{1+xy}\, j$$

At $P(3,1)$:

$$\nabla f(3,1) = \frac{3(1)}{2}\sqrt{1+3}\, i + \frac{3(3)}{2}\sqrt{1+3}\, j = 3i + 9j$$

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- Directional Derivative:

$$D_u f = (3i + 9j) \cdot \left(\frac{1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j\right) = \frac{3}{\sqrt{2}} + \frac{9}{\sqrt{2}} = 6\sqrt{2}$$ Ans

2. $f(x,y) = \sin(5x - 3y)$; $P(3,5)$; $u = \frac{3}{5} i - \frac{4}{5} j$

Solve

$$\nabla f(x,y) = 5\cos(5x-3y) i - 3\cos(5x-3y) j$$

At $P(3,5)$:

$$\nabla f(3,5) = 5\cos(0) i - 3\cos(0) j = 5i - 3j$$

Directional Derivative:

$$D_u f = (5i - 3j) \cdot \left(\frac{3}{5} i - \frac{4}{5} j\right) = 3 + \frac{12}{5} = \frac{27}{5}$$ Ans

3. $f(x,y) = \ln(1 + x^2 + y)$; $P(0,0)$; $u = -\frac{1}{\sqrt{10}} i - \frac{3}{\sqrt{10}} j$

- Gradient:

$$\nabla f(x,y) = \frac{2x}{1+x^2+y} i + \frac{1}{1+x^2+y} j$$

$$\nabla f(0,0) = \frac{0}{1} i + \frac{1}{1} j = j$$

$$D_u f = \nabla f \cdot \hat{u}$$

$$D_u f = (0i + 1j) \cdot \left(-\frac{1}{\sqrt{10}} i - \frac{3}{\sqrt{10}} j\right)$$

$$= 0 - \frac{3}{\sqrt{10}} = -\frac{3}{\sqrt{10}}$$ Ans

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Date:

4. $f(x,y) = \frac{cx+dy}{x-d}$; $P(3,4)$; $u = \frac{4}{5}i + \frac{3}{5}j$

$$\nabla f = f_x \hat{i} + f_y \hat{j}$$

$$\nabla f = \frac{\partial}{\partial x}\left(\frac{cx+dy}{x-d}\right)i + \frac{\partial}{\partial y}\left(\frac{cx+dy}{x-y}\right)j$$

$$\nabla f = \left(\frac{c(x-y)-(cx+dy)(1)}{(x-y)^2}\right)i + \left(\frac{d(x-y)-(cx+dy)(-1)}{(x-y)^2}\right)j$$

$$\nabla f = \left(\frac{-(c+d)y}{(x-y)^2}\right)i + \left(\frac{(c+d)x}{(x-y)^2}\right)j$$

At $P(3,4)$:

$$\nabla f(3,4) = \left(\frac{-(c+d)4}{(3-4)^2}\right)i + \left(\frac{(c+d)3}{(3-4)^2}\right)j$$

$$= -4(c+d)i + 3(c+d)j$$

Directional Derivative:

$$D_u f = (-4(c+d)i + 3(c+d)j) \cdot \left(\frac{4}{5}i + \frac{3}{5}j\right)$$

$$D_u f = -\frac{16}{5}(c+d) + \frac{9}{5}(c+d) = -\frac{7}{5}(c+d)$$

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Date:

5. $f(x,y,z) = 4x^5y^2z^3$; $P(2,-1,1)$; $u = \frac{1}{3}i + \frac{2}{3}j - \frac{2}{3}k$

$$\nabla f = \frac{\partial}{\partial x}(4x^5y^2z^3)i + \frac{\partial}{\partial y}(4x^5y^2z^3)j + \frac{\partial}{\partial z}(4x^5y^2z^3)k$$

$$\nabla f = (20x^4y^2z^3)i + (8x^5yz^3)j + (12x^5y^2z^2)k$$

At $P(2,-1,1)$:

$$\nabla f(2,-1,1) = (20(2)^4(-1)^2(1)^3)i + (8(2)^5(-1)(1)^3)j + (12(2)^5(-1)^2(1)^2)k$$

$$= 320i - 256j + 384k$$

Directional Derivative:

$$D_u f = (320i - 256j + 384k) \cdot \left(\frac{1}{3}i + \frac{2}{3}j - \frac{2}{3}k\right)$$

$$D_u f = \frac{320}{3} - \frac{512}{3} - \frac{768}{3} = -\frac{960}{3} = -320$$

6. $f(x,y,z) = ye^{xz} + z^2$; $P(0,2,3)$; $u = \frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k$

$$\nabla f = \frac{\partial}{\partial x}(ye^{xz} + z^2)i + \frac{\partial}{\partial y}(ye^{xz} + z^2)j + \frac{\partial}{\partial z}(ye^{xz} + z^2)k$$

$$= (yze^{xz})i + (e^{xz})j + (xye^{xz} + 2z)k$$

At $P(0,2,3)$:

$$\nabla f(0,2,3) = ((2)(3)e^0)i + (e^0)j + ((0)(2)e^0 + 2(3))k$$

$$\nabla f(0,2,3) = 6i + j + 6k.$$

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Date:

Directional Derivative:

$$D_u f = (6i + j + 6k) \cdot \left(\frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k\right)$$

$$D_u f = \frac{12}{7} - \frac{3}{7} + \frac{36}{7} = \frac{45}{7}$$ Ans

7. $f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2)$; $P(-1, 2, 4)$; $u = -\frac{3}{13}i - \frac{4}{13}j - \frac{12}{13}k$.

$$\nabla f = \left(\frac{2x}{x^2+2y^2+3z^2}\right)i + \left(\frac{4y}{x^2+2y^2+3z^2}\right)j + \left(\frac{6z}{x^2+2y^2+3z^2}\right)k$$

At $P(-1, 2, 4)$:

Denominator is $(-1)^2 + (2)^2_2 + 3(4)^2 = 1 + 8 + 48 = 57$.

$$\nabla f(-1, 2, 4) = \left(\frac{2(-1)}{57}\right)i + \left(\frac{4(2)}{57}\right)j + \left(\frac{6(4)}{57}\right)k$$

$$= -\frac{2}{57}i + \frac{8}{87}j + \frac{24}{57}k$$

Directional Derivative:

$$D_u f = \left(-\frac{2}{57}i + \frac{8}{57}j + \frac{24}{57}k\right) \cdot \left(-\frac{3}{13}i - \frac{4}{13}j - \frac{12}{13}k\right)$$

$$D_u f = \frac{6}{741} - \frac{32}{741} - \frac{288}{741} = -\frac{314}{741}$$ Ans

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8. $f(x,y,z) = \sin xyz$; $P\left(\frac{1}{2}, \frac{1}{3}, \pi\right)$; $u = \frac{1}{\sqrt{3}}i - \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k$.

$$\nabla f = (yz\cos(xyz))i + (xz\cos(xyz))j + (xy\cos(xyz))k$$

At $P(1/2, 1/3, \pi)$:

The argument is $xyz = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)(\pi) = \frac{\pi}{6}$, so $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$.

$$\nabla f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) = \left(\left(\frac{1}{3}\right)(\pi)\frac{\sqrt{3}}{2}\right)i + \left(\left(\frac{1}{2}\right)(\pi)\frac{\sqrt{3}}{2}\right)j + \left(\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)\frac{\sqrt{3}}{2}\right)k$$

$$= \frac{\pi\sqrt{3}}{6}i + \frac{\pi\sqrt{3}}{4}j + \frac{\sqrt{3}}{12}k.$$

Directional Derivative:

$$D_u f = \left(\frac{\pi\sqrt{3}}{6}i + \frac{\pi\sqrt{3}}{4}j + \frac{\sqrt{3}}{12}k\right) \cdot \left(\frac{1}{\sqrt{3}}i - \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k\right)$$

$$D_u f = \left(\frac{\pi\sqrt{3}}{6}\right)\left(\frac{1}{\sqrt{3}}\right) + \left(\frac{\pi\sqrt{3}}{4}\right)\left(\frac{-1}{\sqrt{3}}\right) + \left(\frac{\sqrt{3}}{12}\right)\left(\frac{1}{\sqrt{3}}\right)$$

$$D_u f = \frac{\pi}{6} - \frac{\pi}{4} + \frac{1}{12} = \frac{2\pi - 3\pi + 1}{12}$$

$$D_u f = \frac{1 - \pi}{12}$$

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9-18 Find the directional derivative of f at P in the direction of a.

9. $f(x,y) = 4x^3y^2$; $P(2,1)$; $a = 4i - 3j$

1. Find the unit vector (u).

First find the magnitude of a:

$$|a| = \sqrt{4^2 + (-3)^2} = \sqrt{16+9} = \sqrt{25} = 5$$

Now, create the unit vector u:

$$u = \frac{a}{|a|} = \frac{4i - 3j}{5} = \frac{4}{5}i - \frac{3}{5}j$$

2. Find the Gradient (∇f).

$$\nabla f = (12x^2y^2)i + (8x^3y)j$$

$$\nabla f(2,1) = (12(2)^2(1)^2)i + (8(2)^3(1))j = 48i + 64j$$

3. Find the directional Derivative ($D_u f$).

$$D_u f = \nabla f(2,1) \cdot \hat{u}$$

$$= (48i + 64j) \cdot \left(\frac{4}{5}i - \frac{3}{5}j\right)$$

$$= (48)\left(\frac{4}{5}\right) + (64)\left(\frac{-3}{5}\right) = \frac{192}{5} - \frac{192}{5}$$

$$= 0$$

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10. $f(x,y) = 9x^3 - 2y^3$; $P(1,0)$; $a = i - j$

Solve

1. $|a| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$

$u = \frac{a}{|a|} = \frac{i - j}{\sqrt{2}} = \frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j$

$\nabla f = (27x^2) i - (6y^2) j$

At $P(1,0)$: $\nabla f(1,0) = (27(1)^2) i - (6(0)^2) j = 27i$

Directional Derivative ($D_u f$):

$D_u f = (27i) \cdot \left(\frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j \right) = \frac{27}{\sqrt{2}}$ Ans

11. $f(x,y) = y^2 \ln x$; $P(1,4)$; $a = -3i + 3j$

Solve

$|a| = \sqrt{(-3)^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$

$u = \frac{a}{|a|} = \frac{-3i + 3j}{3\sqrt{2}} = \frac{-1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$

$\nabla f = \left(\frac{y^2}{x} \right) i + (2y \ln x) j$

$\nabla f(1,4) = \left(\frac{4^2}{1} \right) i + (2(4) \ln 1) j = 16i$

Directional Derivative ($D_u f$):

$D_u f = (16i) \cdot \left(\frac{-1}{\sqrt{2}} i + \frac{1}{2} j \right)$

$= \frac{-16}{\sqrt{2}} = -8\sqrt{2}$ Ans

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12. $f(x,y) = e^x \cos y$; $P(0, \pi/4)$; $a = 5i - 2j$

solve

$$|a| = \sqrt{5^2 + (-2)^2} = \sqrt{29}, \quad u = \frac{5i - 2j}{\sqrt{29}} = \frac{5}{\sqrt{29}} i - \frac{2}{\sqrt{29}} j$$

$$\nabla f = (e^x \cos y) i - (e^x \sin y) j$$

At $P(0, \pi/4)$:

$$\nabla f(0, \pi/4) = \left(e^0 \cos\left(\frac{\pi}{4}\right)\right) i - \left(e^0 \sin\left(\frac{\pi}{4}\right)\right) j$$

$$= \frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j$$

Directional Derivative:

$$D_u f = \left(\frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j\right) \cdot \left(\frac{5}{\sqrt{29}} i - \frac{2}{\sqrt{29}} j\right)$$

$$= \frac{5}{\sqrt{58}} + \frac{2}{\sqrt{58}} = \frac{7}{\sqrt{58}}$$ ans

13. $f(x,y) = \tan^{-1}\left(\frac{y}{x}\right)$; $P(-2,2)$; $a = -i - j$

solve

$$|a| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$u = \frac{a}{|a|} = \frac{-i - j}{\sqrt{2}} = -\frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j$$

$$\nabla f = \left(\frac{1}{1 + (y/x)^2} \cdot \frac{-y}{x^2}\right) i + \left(\frac{1}{1 + (y/x)^2} \cdot \frac{1}{x}\right) j$$

$$= \left(\frac{-y}{x^2 + y^2}\right) i + \left(\frac{x}{x^2 + y^2}\right) j$$

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At $P(-2, 2)$:

$$\nabla f(-2,2) = \left(\frac{-2}{(-2)^2+2^2}\right)i + \left(\frac{-2}{(-2)^2+2^2}\right)j$$

$$= \frac{-2}{8_4}i - \frac{2}{8_4}j = -\frac{1}{4}i - \frac{1}{4}j$$

Directional Derivative:

$$D_u f = \left(-\frac{1}{4}i - \frac{1}{4}j\right) \cdot \left(\frac{-1}{\sqrt{2}}i - \frac{1}{\sqrt{2}}j\right)$$

$$= \frac{1}{4\sqrt{2}} + \frac{1}{4\sqrt{2}} = \frac{2}{4\sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$$ dw

14. $f(x,y) = xe^y - ye^x$; $P(0,0)$; $a = 5i - 2j$ dw

$$|a| = \sqrt{5^2 + (-2)^2} = \sqrt{29}$$

$$u = \frac{5i - 2j}{\sqrt{29}}$$

$$\nabla f = (e^y - ye^x)i + (xe^y - e^x)j$$

At $P(0,0)$:

$$\nabla f(0,0) = (e^0 - 0e^0)i + (0e^0 - e^0)j = i - j$$

Directional Derivative:

$$D_u f = (i - j) \cdot \left(\frac{5}{\sqrt{29}}i - \frac{2}{\sqrt{29}}j\right)$$

$$= \frac{5}{\sqrt{29}} + \frac{2}{\sqrt{29}} = \frac{7}{\sqrt{29}}$$ dw

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15. $f(x,y,z) = xy + z^2$; $P(-3,0,4)$; $a = i + j + k$.

$|a| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$u = \frac{a}{|a|} = \frac{i+j+k}{\sqrt{3}} = \frac{1}{\sqrt{3}}i + \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k$

$\nabla f = yi + xj + 2z\,k$

At $P(-3,0,4)$:

$\nabla f(-3,0,4) = (0)i + (-3)j + 2(4)k = -3j + 8k$

Directional Derivative:

$D_u f = (-3j + 8k) \cdot \left(\frac{1}{\sqrt{3}}i + \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k\right)$

$= -\frac{3}{\sqrt{3}} + \frac{8}{\sqrt{3}} = \frac{5}{\sqrt{3}}$

17. $f(x,y,z) = \frac{z-x}{z+y}$; $P(1,0,-3)$; $a = -6i + 3j - 2k$

$|a| = \sqrt{(-6)^2 + (3)^2 + (-2)^2} = \sqrt{49} = 7$

$u = \frac{a}{|a|} = \frac{-6i+3j-2k}{7} = -\frac{6}{7}i + \frac{3}{7}j - \frac{2}{7}k$

$\nabla f = \left(-\frac{1}{z+y}\right)i - \left(\frac{z-x}{(z+y)^2}\right)j + \left(\frac{x+y}{(z+y)^2}\right)k$

At $P(1,0,-3)$:

$\nabla f(1,0,-3) = \left(-\frac{1}{-3+0}\right)i - \left(\frac{-3-1}{(-3+0)^2}\right)j + \left(\frac{1+0}{(-3+0)^2}\right)k$

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$$= \frac{1}{3}i + \frac{4}{9}j + \frac{1}{9}k$$

Directional Derivative:

$$D_u f = \left(\frac{1}{3}i + \frac{4}{9}j + \frac{1}{9}k\right) \cdot \left(\frac{-6}{7}i + \frac{3}{7}j - \frac{2}{7}k\right)$$

$$= -\frac{6}{21} + \frac{12}{63} - \frac{2}{63} = \frac{-18}{63} + \frac{10}{63} = \frac{-8}{63}$$ Ans

16. $f(x,y,z) = y - \sqrt{x^2+z^2}$; $P(-3,1,4)$; $a = 2i - 2j - k$.

Solve

$$|a| = \sqrt{2^2 + (-2)^2 + (-1)^2} = \sqrt{9} = 3$$

$$u = \frac{a}{|a|} = \frac{2i - 2j - k}{3} = \frac{2}{3}i - \frac{2}{3}j - \frac{1}{3}k$$

$$\nabla f = \left(-\frac{x}{\sqrt{x^2+z^2}}\right)i + (1)j - \left(\frac{z}{\sqrt{x^2+z^2}}\right)k$$

At $P(-3,1,4)$:

$$\nabla f(-3,1,4) = \left(-\frac{-3}{\sqrt{(-3)^2+4^2}}\right)i + j - \left(\frac{4}{\sqrt{(-3)^2+4^2}}\right)k$$

$$= \frac{3}{5}i + j - \frac{4}{5}k$$

Directional Derivatives:

$$D_u f = \left(\frac{5}{3}i + j - \frac{4}{5}k\right) \cdot \left(\frac{2}{3}i - \frac{2}{3}j - \frac{1}{3}k\right)$$

$$= \frac{6}{15} - \frac{2}{3} + \frac{4}{15} = \frac{10}{15} - \frac{10}{15} = 0$$ Ans

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18. $f(x,y,z) = e^{x+y+3z}$; $P(-2,2,-1)$; $a = 20i - 4j + 5k$

$|a| = \sqrt{(20)^2 + (-4)^2 + (5)^2} = \sqrt{441} = 21$

$u = \frac{a}{|a|} = \frac{20}{21}i - \frac{4}{21}j + \frac{5}{21}k$

$\nabla f = f_x i + f_y j + f_z k$

$= e^{x+y+3z} i + e^{x+y+3z} j + 3e^{x+y+3z} k = e^{x+y+3z}(i+j+$

At $P(-2,2,-1)$:

$\nabla f(-2,2,-1) = e^{-2+2+3(-1)}(i+j+3k) = e^{-3}(i+j+3k)$

Directional Derivative:

$D_u f = (e^{-3}(i+j+3k)) \cdot \left(\frac{20}{21}i - \frac{4}{21}j + \frac{5}{21}k\right)$

$= \frac{e^{-3}}{21}(1(20) + 1(-4) + 3(5))$

$= \frac{e^{-3}}{21}(20 - 4 + 15) = \frac{31}{21}e^{-3}$

33-40 Find ∇z or ∇w

33. $z = \sin(7y^2 - 7xy)$

$\nabla z = \left(-7y\cos(7y^2 - 7xy)\right)i + \left((14y - 7x)\cdot\cos(7y^2 - 7xy)\right)j$

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34 $z = 7 \sin \sin (6x/y)$

sol:

$$\nabla z = 7\cos\left(\frac{6y}{y}\right)\cdot\left(\frac{6}{y}\right)i + 7\cos\left(\frac{6x}{y}\right)\cdot\left(\frac{-6x}{y^2}\right)j$$

$$= \frac{42}{y}\cos\left(\frac{6x}{y}\right)i - \frac{42x}{y^2}\cos\left(\frac{6x}{y^2}\right)j$$ Ans

35. $z = \frac{6y+7y}{6x-7y}$

sol:

$$\nabla z = \frac{6(6x-7y)-(6x+7y)(6)}{(6x-7y)^2}i + \frac{7(6x-7y)-(6x+7y)(-7)}{(6x-7y)^2}$$

$$= \frac{36x-42y-36x-42y}{(6x-7y)^2}i + \frac{42x-49y+42x+49y}{(6x-7y)^2}j$$

$$= -\frac{84y}{(6x-7y)^2}i + \frac{84x}{(6x-7y)^2}j$$ Ans

36. $z = \frac{6xe^{3y}}{x+8y}$

sol:

$$\nabla z = \frac{6e^{3y}(x+8y)-6xe^{3y}(1)}{(x+8y)^2}i + \frac{18xe^{3y}(x+8y)-6xe^{3y}(8)}{(x+8y)^2}j$$

$$= \frac{48ye^{3y}}{(x+8y)^2}i + \frac{18x^2e^{3y}+144xye^{3y}-48xe^{3y}}{(x+8y)^2}j$$

$$= \frac{48ye^{3y}}{(x+8y)^2}i + \frac{6xe^{3y}(3x+24y-8)}{(x+8y)^2}j$$ Ans

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37. $w = -x^9 - y^3 + z^{12}$

solve

$$\nabla w = \frac{\partial w}{\partial x} i + \frac{\partial w}{\partial y} j + \frac{\partial w}{\partial z} k$$

$$= 9x^8 i - 3y^2 j + 12z^{11} k$$

38. $w = x e^{8y} \sin(6z)$

solve

$$\nabla w = \frac{\partial w}{\partial x} i + \frac{\partial w}{\partial y} j + \frac{\partial w}{\partial z} k$$

$$= e^{8y} \sin(6z) i + 8x e^{8y} \sin(6z) j + 6x e^{8y} \cos(6z) k$$

39. $w = \ln \sqrt{x^2 + y^2 + z^2} = \frac{1}{2} \ln(x^2 + y^2 + z^2)$

solve

$$\nabla w = \frac{\partial w}{\partial x} i + \frac{\partial w}{\partial y} j + \frac{\partial w}{\partial z} k$$

$$= \frac{1}{2}\left(\frac{2x}{x^2+y^2+z^2}\right) i + \frac{1}{2}\left(\frac{2y}{x^2+y^2+z^2}\right) j + \frac{1}{2}\left(\frac{2z}{x^2+y^2+z^2}\right) k$$

$$= \frac{x}{x^2+y^2+z^2} i + \frac{y}{x^2+y^2+z^2} j + \frac{z}{x^2+y^2+z^2} k$$

40. $w = e^{-5x} \sec(x^2 yz)$

solve

$$\nabla w = \frac{\partial w}{\partial x} i + \frac{\partial w}{\partial y} j + \frac{\partial w}{\partial z} k$$

$\partial w/\partial x = -5e^{-5x} \sec(x^2yz) + e^{-5x} \sec(x^2yz) \tan(x^2yz) \cdot (2xyz)$

$\partial w/\partial y = e^{-5x} \sec(x^2yz) \tan(x^2yz) \cdot (x^2z)$

$\partial w/\partial z = e^{-5x} \sec(x^2yz) \tan(x^2yz) \cdot (x^2y)$

$\nabla w = e^{-5x} \sec(x^2yz)[(2xyz \tan(x^2yz) - 5) i + x^2z \tan(x^2yz) j + x^2y \tan(x^2yz) k]$

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Find the gradient of f at the indicated point.

41. $f(x,y) = 5x^2 + y^4 ; (4,2)$

Solve

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j = 10x\,i + 4y^3 j$$

$$\nabla f(4,2) = 10(4)\,i + 4(2)^3 j = 40i + 32j \quad \text{Ans}$$

42. $f(x,y) = 5\sin x^2 + \cos 3y ; (\sqrt{\pi}/2, 0)$

Solve

$$\nabla f = 5\cos(x^2)(2x)i + (-\sin(3y)(3)).$$
$$= 10x\cos(x^2)i - 3\sin(3y)j$$

$$\nabla f(\sqrt{\pi}/2, 0) = 10\left(\frac{\sqrt{\pi}}{2}\right)\cos\left(\frac{\pi}{4}\right)i - 3\sin(0)j$$

$$= 5\sqrt{\pi}\left(\frac{\sqrt{2}}{2}\right) - 0 = \frac{5\sqrt{2\pi}}{2}\,i \quad \text{Ans}$$

43. $f(x,y) = (x^2 + xy)^3 ; (-1, -1)$

Solve

$$\nabla f = 3(x^2+xy)^2(2x+y)i + 3(x^2+xy)^2(x)j$$

$$\nabla f(-1,-1) = 3((-1)^2 + (-1)(-1))^2(2(-1)+(-1))i + 3((-1)^2 + (-1)(-1))^2 (-1)j$$

$$= 3(2)^2(-3)i + 3(2)^2(-1)j$$
$$= 12(-3)i + 12(-1)j$$
$$= -36i - 12j \quad \text{Ans}$$

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Date:

44. $f(x,y) = (x^2+y^2)^{-\frac{1}{2}}$; $(3,4)$

$$\nabla f = -x(x^2+y^2)^{-\frac{3}{2}} i - y(x^2+y^2)^{-3/2} j$$

At $(3,4)$: $x^2+y^2 = 3^2+4^2 = 25$, so $(x^2+y^2)^{-\frac{3}{2}} = (25)^{-\frac{3}{2}} = \frac{1}{125}$

$$\nabla f(3,4) = -3\left(\frac{1}{125}\right) i - 4\left(\frac{1}{125}\right) j$$

$$= -\frac{3}{125} i - \frac{4}{125} j$$

Ex # 13.7

Tangent Plane & Normal Line :-

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$a = f_x$
$b = f_y$, at (x_0, y_0, z_0)
$c = f_z$

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$$

3-12 Find an equation for the tangent plane and parametric equations for the normal line to the surface at the point P.

3. $x^2 + y^2 + z^2 = 25$; $P(-3, 0, 4)$

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